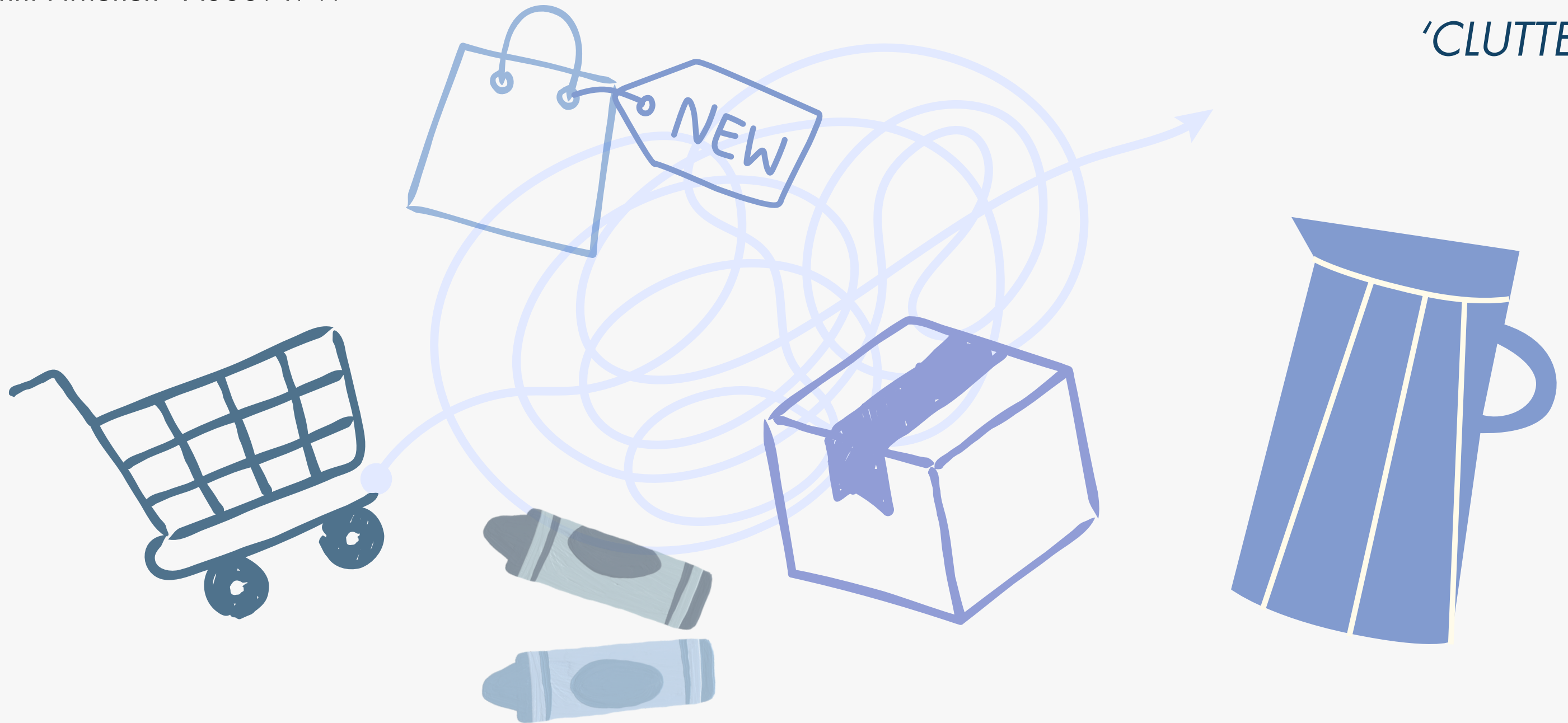


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# *THE ACCUMULATION OF UNUSED HOUSEHOLD ITEMS: 'CLUTTER'*



**PBL202**  
**PROBLEM BASED LEARNING**

**PROJECT ROUND**  
**3: RESURFACE**

# INDEX

|                                       |      |
|---------------------------------------|------|
| GROUP                                 | 3    |
| PREMISE                               | 4    |
| PROBLEM DEFINITION                    | 5    |
| CONTEXT & RESEARCH INSIGHTS           | 6    |
| STAKEHOLDER MAPS                      | 7,8  |
| SCRAPPY PROPTYPES                     | 9,10 |
| PROBLEM REFRAMING                     | 11   |
| BRAINSTORMING                         | 12   |
| EVORATIVE PROTOTYPE                   | 13   |
| HUMAN BASED RESEARCH                  | 14   |
| PROTOTYPE TESTING & EVALUATION        | 15   |
| NEXT STEPS                            | 16   |
| USER STORY / JOURNEY / BEFORE & AFTER | 17   |
| WHY THIS WORKS                        | 18   |
| SOLUTION IN USE                       | 19   |
| SYSTEM IMPACT                         | 20   |
| LIMITATIONS & FUTURE DIRECTIONS       | 21   |
| REFLECTION & CLOSING STATEMENT        | 22   |
| REFERENCES                            | 23   |

“WE HAVE  
TOO MUCH  
STUFF AND  
NOT ENOUGH  
HAPPINESS.”

- Annie Leonard ;  
an environmental activist and filmmaker known for exposing how  
modern consumer systems drive overconsumption and waste.

# GROUP

Our team brings together three complementary design disciplines, enabling a balanced and interdisciplinary approach to this project. A group agreement established shared expectations around communication, deadlines, and accountability from the outset.

## Chloe Tjoa

### Visual Communication Designer

Chloe brings expertise in visual systems, layout, and information design. She led the system and stakeholder mapping, prototype coordination, ideation process, document design, and ensured visual consistency across all deliverables.

## Kira Zakirov

### Design Researcher & Strategist

Kira's background spans critical thinking, critical writing, and strategic framing. She led report drafting and finishing, reference management, ideation process, and the synthesis of research insights into actionable design directions.

## Sanni Pirhonen

### Visual Designer & Research Specialist

Sanni brings prior academic and practical experience in sustainability research and circular systems. Her practice focuses on translating complex ideas into coherent visual narratives. She led the visual design of the document, research coordination, ideation process, and grounded the project's direction in evidence-based thinking.

Despite defined roles, all members contributed equally to discussions, decision-making, and weekly iterations. The team operated through a design thinking framework, with regular check-ins to respond to emerging insights and refine the project direction.

# PREMISE

This project aims to investigate the systematic drivers of household accumulation and identify design-led interventions that reduce redistribution friction.

Through system and stakeholder mapping, contextual research, and iterative prototyping, the group will explore how behavioral and structural shifts may rebalance the flow of goods within domestic environments.



**70%** of donated items  
can't be absorbed by  
op-shops (SBS,2025)

**30 - 50** year-olds  
carry the highest  
accumulation burden

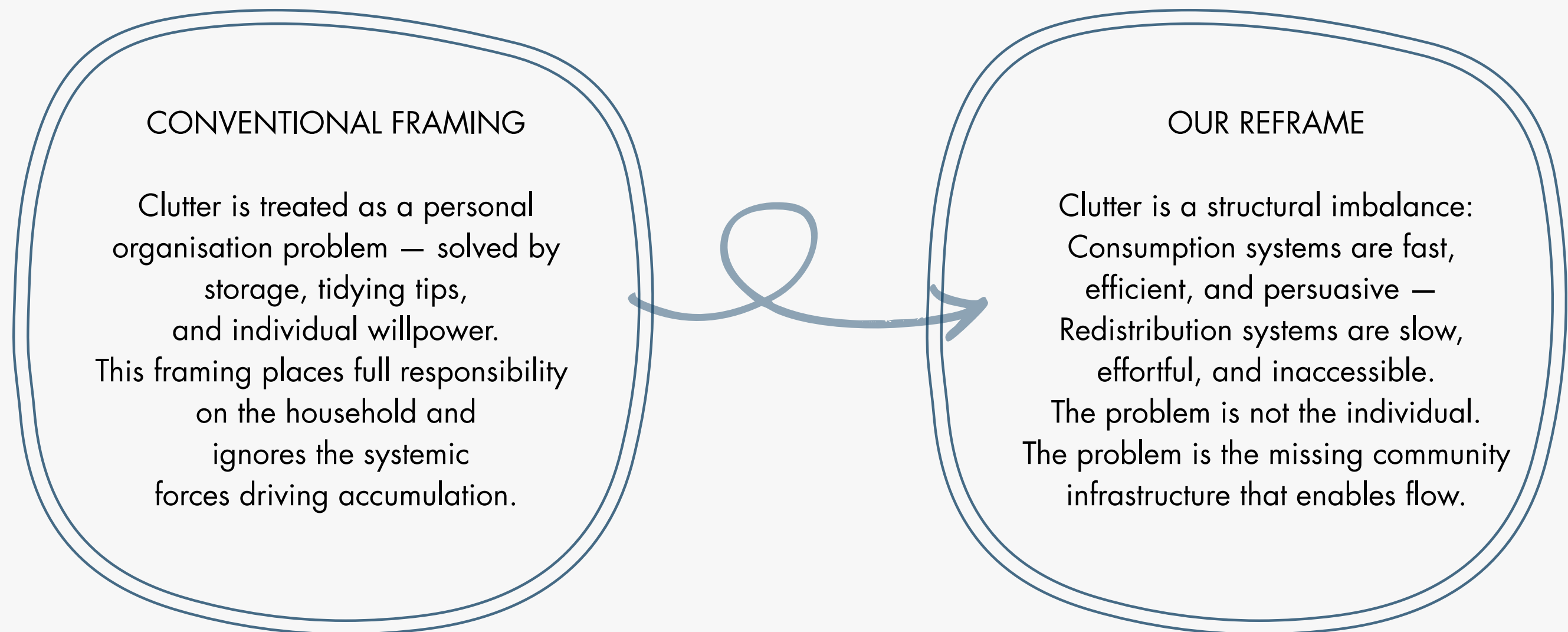
**Loneliness &**  
lack of belonging  
accelerate clutter cycles

*Australia's overshoot day, calculated by Global Footprint Network, falls around late March each year, meaning the country uses up its annual ecological resources in just under three months.*

# PROBLEM DEFINITION

Household clutter stems from a systemic imbalance rather than individual behaviour alone. While retail systems are optimised for speed and accessibility, redistribution pathways—donation, resale, repair, and recycling—are fragmented and time-intensive. As a result, items enter homes more easily than they exit, leading to accumulation.

Psychological barriers such as decision fatigue and perceived future value further delay action. As Glawson (2021) notes, consumer culture normalises continuous acquisition, reinforcing inaction. Consequently, clutter reflects a broader failure of circular systems, contributing to household strain and environmental impacts, including textile waste overflow in regions such as Ghana (Quashie-Idun, 2025).



# CONTEXT & RESEARCH INSIGHTS



## OVERCONSUMPTION & FAST RETAIL

- Fast retail cycles and influencer marketing drive high purchasing frequency, flooding homes faster than goods can flow out (Cooper, 2005; Dittmar, 2008).



## PSYCHOLOGICAL BARRIERS

- Emotional attachment, 'just in case' mindset, and decision fatigue delay disposal. Clutter triggers overwhelm, leading to avoidance and further accumulation (Belk, 1988).



## REDISTRIBUTION FRICTION

- Donation systems are selective and time-consuming. Resale platforms require effort. Over 70% of donated textiles cannot be absorbed by op shops and are exported (SBS Australia, 2025).



## LONELINESS & COMMUNITY LOSS

- People aging alone accumulate clutter as a coping mechanism. A sense of belonging and shared purpose is a known protective factor (Jackson, 2017).



## ENVIRONMENTAL SYSTEMIC COST

- Excess goods ending up in Global South waste sites (e.g., Ghana) reveal the environmental cost of domestic accumulation (Quashie-Idun, 2025).



## COMMUNITY AS INFRASTRUCTURE

- Research on sharing economies shows that low-barrier, local exchange systems increase participation in circular consumption (Ellen MacArthur Foundation, 2013).



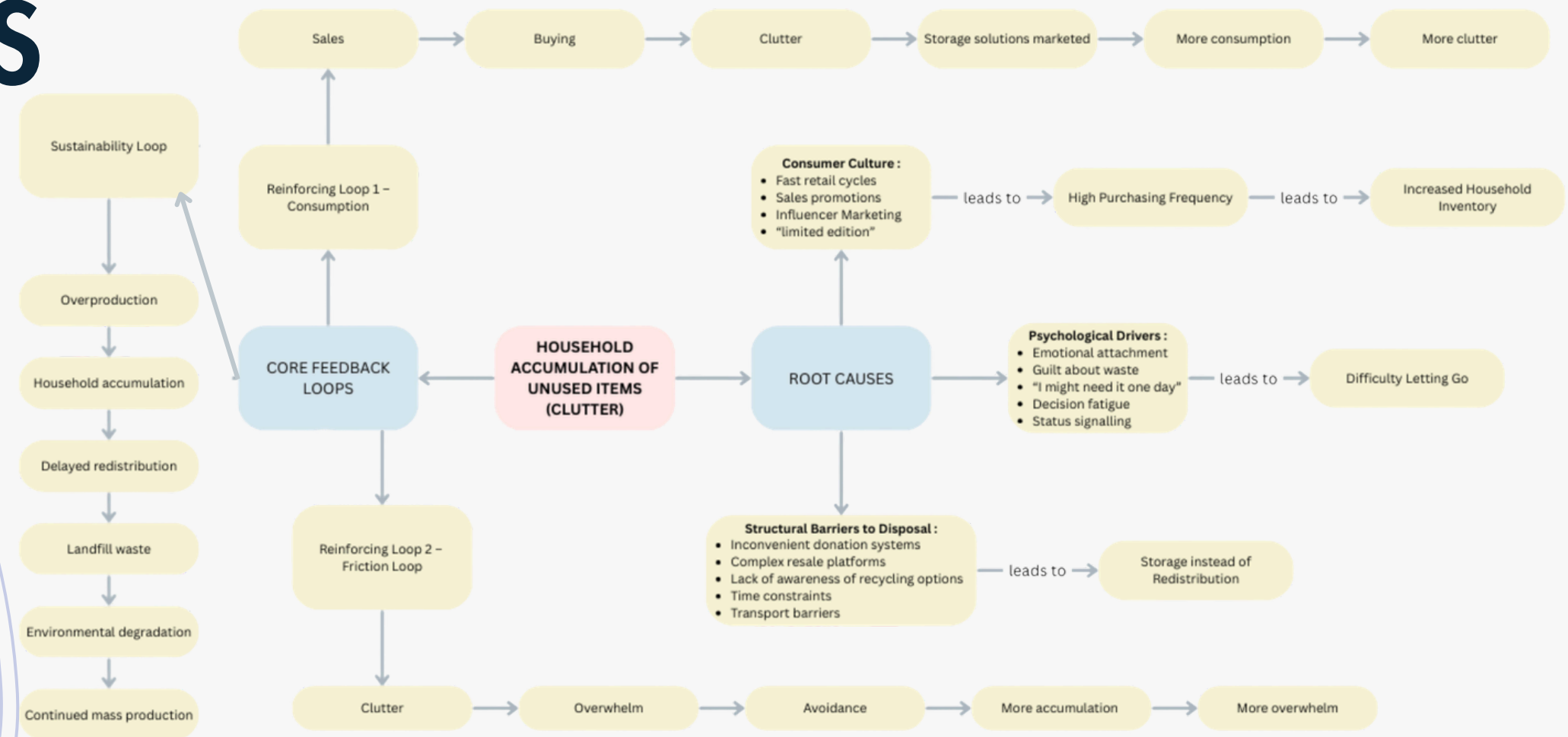
# SYSTEM MAPS

## PRIMARY

Directly Affected

- Young renters (18–35)
- Families with children
- Students
- Elderly residents (inner north)
- First-gen immigrants

Impact: mental stress, reduced space, financial waste



## SECONDARY

Indirectly Involved

- Local charities & thrift stores
- Online resale platforms (eBay, FB Marketplace)
- Local council waste services
- Community centres

Role: redistribution channels — often overwhelmed

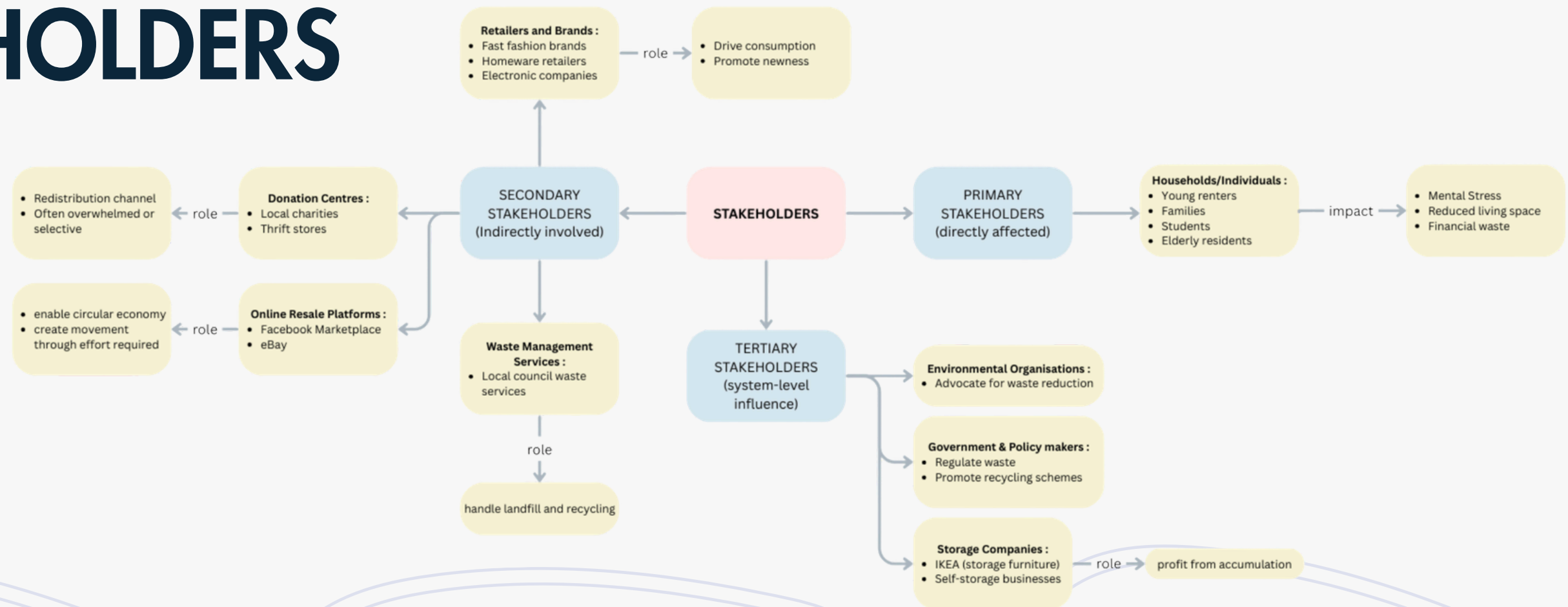
## TERTIARY

System-Level Influence

- Environmental organisations
- Government & policymakers
- Retailers & fast fashion brands
- Storage companies (IKEA et al.)

Role: systemic drivers & regulatory levers

# STAKEHOLDERS MAPS



## PRIMARY THE OVERWHELMED HOUSEHOLD *Directly Affected*

- Interest: a comfortable, clutter-free home without spending excessive time or money on redistribution
- Pain: no clear, trusted, low-effort pathway to let go of unused items > donation centres are selective, resale platforms require effort, decision fatigue prevents action
- Influence on problem: the direct source of accumulated goods > behavioural change
- at this level is the primary lever for reducing household clutter

## SECONDARY DONATION CENTRES & THRIFT STORES *Indirectly Involved*

- Interest: to receive quality, usable donations that fund community programmes and support people in need
- Pain: structurally overwhelmed > over 70% of donated textiles cannot be absorbed and must be exported; limited storage, staff, and sorting capacity force selective acceptance
- Influence on problem: the main redistribution channel in the system > when this node fails, goods bypass circulation entirely and go straight to landfill

## TERTIARY GOVERNMENT & POLICYMAKERS *System-Level Influence*

- Interest: to reduce landfill pressure, promote recycling schemes, and meet sustainability targets at a municipal and national level
- Pain: policy instruments are slow and blunt > awareness campaigns alone do not change behaviour; infrastructure investment is costly and politically deprioritised
- Influence on problem: holds the regulatory lever > extended producer responsibility laws, subsidised redistribution infrastructure, and circular economy mandates could structurally shift the system



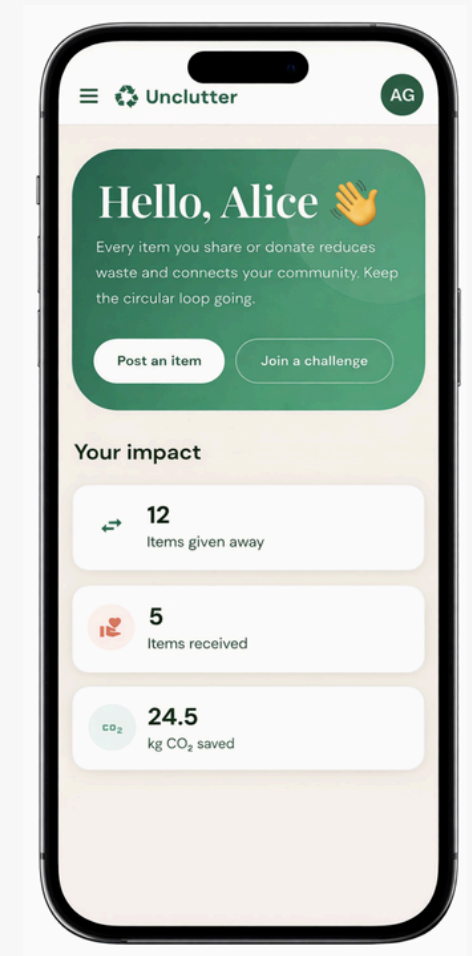
# SCRAPPY PROTOTYPE(S)



P2 **The Neighborhood Swap Hub** is a community-based redistribution model designed to reduce household clutter by making it easier for people to exchange items locally. Tests local social redistribution problems.

P3 **The Circular Route Van** is a mobile redistribution service designed to reduce household clutter by bringing donation, resale, and recycling opportunities directly into neighborhoods. Tests logistical access to redistribution.

P1 **“Unclutter”** is an unpolished **digital prototype** designed to re-balance the flow of goods within households. Tests decision fatigue and low item visibility.



# SCRAPPY PROTOTYPE(S)

P6 **The Declutter Decision Kit** is a simple behavioural mediation tool designed to reduce decision fatigue and lower the problem of redistribution.

1. Printed decision flow chart
2. Five labelled sorting categories:

- Keep
- Repair
- Sell
- Donate
- Recycle



The flowchart guides users through 3 key questions:

- Have I used this in the last 12 months?
- Would I buy this again today?
- Is it easily replaceable if needed?

Based on responses, the item is directed into one of the five categories.

The test decision is overwhelming and unclear.

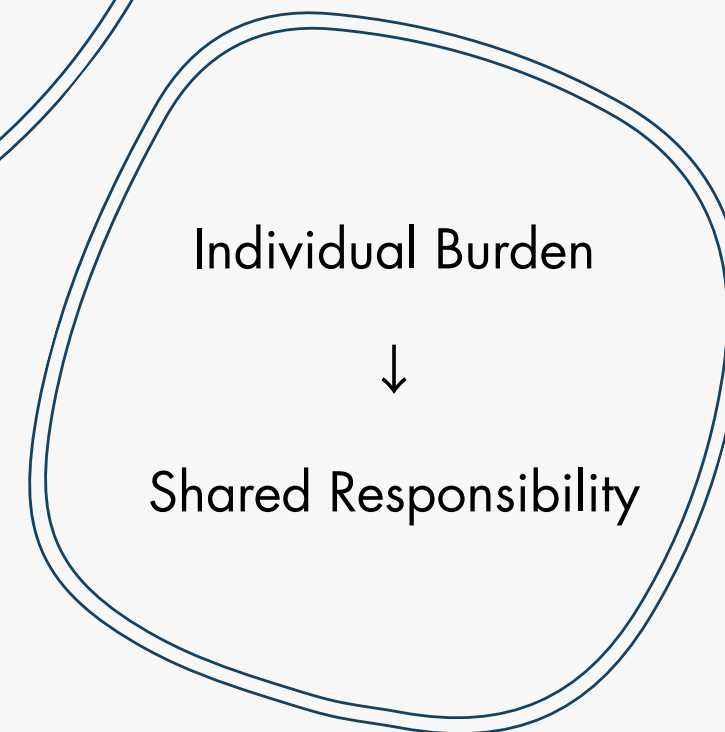
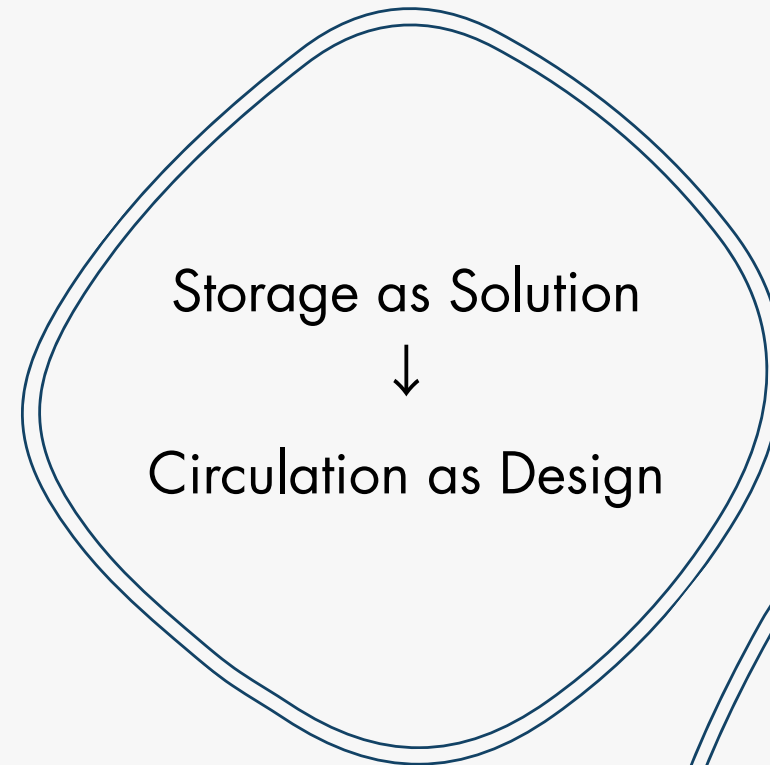
P4 **The Freecycle Community Hub** is a donation centre model focused on the redistribution of usable household items through free access. Tests trust in donation systems.



P5 The “Let It Go” **campaign** is an informational advertising initiative designed to encourage individuals to declutter their homes and redistribute unused items. Tests behavioral awareness and motivation.

# PROBLEM REFRAMING

From Individual Clutter → to Community Disconnection



Collecting clutter and overconsumption may be a symptom of a missing sense of belonging and purpose — a quick-fix dopamine response when community is absent.

The real problem:  
*How do we build community infrastructure that is empathetic, easy, cheap (or free), and low-effort to participate in?*





# BRAINSTORMING

## KEY INSIGHTS THAT SHAPED UNCLUTTER:

Three open brainstorming sessions explored community infrastructure, user behaviour, and platform direction.

These sessions directly informed the Evocative Prototype direction.

- Sharing is caring : people need connection, not just a drop-off point
- Target users: 30–60, immigrants, elderly, retirees, stay-at-home parents
- Clutter as loneliness symptom: replace the dopamine loop with community
- Platform chosen as lowest-barrier entry point; physical events as second layer



# EVOCATIVE PROTOTYPE

## "UNCLUTTER" : COMMUNITY PLATFORM & WEBSITE

Unclutter is a community-first digital platform designed for Melbourne's Inner North, connecting neighbours through sharing and giving ; not as a marketplace, but as a neighbourhood ecosystem built on belonging.

Its direction emerged from prototype testing: the Swap Hub showed social connection drives participation, the Declutter Decision Kit reduced decision fatigue, and the Freecycle Hub built trust through free access. These insights shaped every design decision in Unclutter.

### DESIGN PRINCIPLES

1. Low barrier to participate
2. Community-led, not corporate
3. Free or near-free by Design
4. Focused on Inner North Melbourne
5. Empathetic, not preachy
6. Builds belonging, not guilt



#### GIVE & RECEIVE FEED

Post items to give away. Browse what neighbours are offering. No transactions, no friction. Designed in response to the finding that resale platforms create too much cognitive load for time-poor users.



#### COMMUNITY EVENTS

Swap meets, repair cafés, and skill-sharing sessions hosted locally and discoverable on the platform. People want to know their item goes directly to someone nearby — not into an anonymous warehouse.



#### LOCAL HUB LOCATOR

Find nearest drop-off points, charity partners, and Circular Route Van stops on a live map. Removes the key logistical barrier — people did not know where to take things.



#### IMPACT DASHBOARD

Track your personal contribution: items given, CO<sub>2</sub> saved, connections made. Social motivation — not just tracking — drives engagement.



# HUMAN BASED RESEARCH

Interviews & user insights — Inner North Melbourne, 2025

Semi-structured interviews (n=6) with residents aged 28–58 in Fitzroy, Collingwood and Brunswick. Participants identified as high-accumulation households. Thematic analysis applied.

## THEME 1: *INTENTION ≠ ACTION*

All 6 participants intended to donate or sell items "soon."  
The gap between intention and follow-through was consistently explained by time pressure and not knowing where to take things.

Supports: Lastovicka & Fernandez (2005) — divestment requires a clear, trusted pathway.

## THEME 3: *COMMUNITY AS MOTIVATION*

When asked what would make decluttering feel worthwhile, all 6 cited knowing the item would go directly to someone in their neighbourhood — not an unknown charity warehouse.

Supports: Ellen MacArthur Foundation (2013) — local visibility increases circular participation.

## THEME 2: *SHAME & EMBARRASSMENT*

4 of 6 participants expressed shame around accumulated clutter. Shame was a barrier to seeking help or using public redistribution services.

Supports: Dittmar (2008) — identity and self-concept tied to possessions creates emotional paralysis.

## THEME 4: *DIGITAL FATIGUE*

Older participants (45+) found resale platforms (Facebook Marketplace, eBay) stressful and impersonal. They preferred in-person exchange.

Supports: Cruz-Cárdenas & Arévalo-Chávez (2018) — complexity of disposal channels is a systemic barrier.

# PROTOTYPE TESTING & EVALUATION

## WHAT THE SCRAPPY PROTOTYPES TAUGHT US

| PROTOTYPE              | ASSUMPTION TESTED                | FINDING                                                                    | STATUS       |
|------------------------|----------------------------------|----------------------------------------------------------------------------|--------------|
| Unclutter App          | Low visibility = main barrier    | Confirmed — but users also needed social motivation, not just tracking     | ✓ Validated  |
| Swap Hub               | Convenience drives participation | Strongly confirmed — local + social = highest willingness to participate   | ✓ Validated  |
| Circular Route Van     | Access is the key barrier        | Partially confirmed — access matters, but trust in the system matters more | ~ Partial    |
| Freecycled Hub         | Free = higher donation trust     | Confirmed — non-profit framing significantly increased stated willingness  | ✓ Validated  |
| Let It Go Campaign     | Awareness changes behaviour      | Challenged — awareness alone insufficient; community action needed         | ✗ Challenged |
| Declutter Decision Kit | Decision fatigue is key barrier  | Confirmed — simplified pathways reduced cognitive load significantly       | ✓ Validated  |

# NEXT STEPS

Having completed two rounds of research and prototyping, the following steps outline how this project would continue to evolve toward a fully realised solution.

## 1. EXPAND HUMAN-BASED RESEARCH

Conduct a second round of interviews with a larger and more demographically diverse sample (n=20+), including participants outside Fitzroy, Collingwood, and Brunswick, to test whether findings hold across different community contexts.

## 3. CONDUCT STRUCTURED USABILITY TESTING

Test the Figma prototype with 5–8 participants from the Inner North Melbourne target demographic. Measure task completion rate, emotional response, and perceived trust in the platform.

## 5. RESEARCH SUSTAINABLE GOVERNANCE MODELS

Investigate cooperative and community-owned digital platform structures to ensure Unclutter remains community-led, free to access, and independent from corporate or advertising interests long term.

## 2. DEVELOP A MEDIUM-FIDELITY PROTOTYPE

Move beyond the evocative prototype stage and build a clickable, testable interface in Figma. Priority screens: the Give & Receive Feed, onboarding flow, and Local Hub Locator, based on the highest user need identified in testing.

## 4. MAP COMMUNITY PARTNERSHIP OPPORTUNITIES

Identify and approach potential partners including local councils, op-shops, repair cafés, and community centres in the Inner North to assess feasibility of real-world integration.

# USER STORY / JOURNEY / BEFORE-AFTER

## BEFORE (Without Unclutter)

- Closets overflowing with outgrown kids' clothes and unused kitchen gadgets
- Intends to donate "soon" but never finds time ; bags pile up for months
- Tried Facebook Marketplace twice ; felt stressful, impersonal, gave up
- Feels guilt about waste but shame about the mess prevents asking for help
- Items eventually end up in landfill via council hard rubbish collection



## THE TURNING POINT

- A neighbour mentions Unclutter at school pick-up
- Maya posts 3 bags of kids' clothes on the Give & Receive Feed ; takes under 2 minutes
- Within 48 hours, a family two streets away collects everything
- She receives a message: "These are perfect, my daughter starts school next week"



## AFTER (With Unclutter)

- Maya now regularly posts items she no longer needs ; it feels easy, not overwhelming
- She has picked up a slow cooker and children's books from neighbours
- Joined a local swap meet through the Community Events tab
- Feels connected to her neighbourhood for the first time since moving
- Her home has less clutter , not because she was told to tidy, but because giving became frictionless



**Emotional shift:** Shame → Purpose. Isolation → Belonging. Avoidance → Action.



# WHY THIS WORKS

Unclutter succeeds because it addresses the root causes identified in our research, not the symptoms.

## 1. Reduces Friction, Not Willpower

- Our research found that all 6 participants intended to donate, but were blocked by time pressure and not knowing where to take things (Theme 1: Intention ≠ Action).

Unclutter removes this barrier — posting an item takes under 2 minutes, and local pick-up means no travel.

## 2. Replaces Shame with Social Connection

- 4 of 6 participants expressed shame around clutter (Theme 2).

Unclutter reframes giving as a neighbourly act, not an admission of failure. Community-first design normalises circulation rather than accumulation.

## 3. Leverages Community as Motivation

- All 6 participants said knowing an item goes to a real neighbour makes decluttering worthwhile (Theme 3). Unclutter makes this visible — you see who receives your item and where it goes.

## 4. Addresses Digital Fatigue

- Older participants (45+) found resale platforms stressful (Theme 4).

Unclutter is free, no pricing, no negotiation, no transaction complexity — just give and receive.

## 5. Creates a Positive Feedback Loop

- The Impact Dashboard provides social motivation (items given, CO<sub>2</sub> saved, connections made), reinforcing participation through positive reinforcement rather than guilt.

## 6. Builds Infrastructure, Not Just Awareness

- The "Let It Go" campaign prototype showed awareness alone is insufficient (Challenged status). Unclutter provides the missing infrastructure — the community-level system that makes action possible.

# SOLUTION IN USE

How Unclutter functions in practice — a walkthrough of key features:

## GIVE & RECEIVE FEED

- Post an item with a photo and short description
- Browse what neighbours are offering nearby
  - No pricing, no negotiation — designed for generosity, not transaction
- Directly responds to redistribution friction identified in research

## LOCAL HUB LOCATOR

- Interactive map showing nearest drop-off points, charity partners, and Circular Route Van stops
- Solves the #1 logistical barrier: "I don't know where to take things"
- Integrates existing redistribution infrastructure (op-shops, council services) into one view

## COMMUNITY EVENTS

- Discover local swap meets, repair cafés, and skill-sharing sessions
  - Hosted by community members, surfaced on the platform
- Bridges the digital-physical gap — the platform facilitates real-world connection

## IMPACT DASHBOARD

- Personal metrics: items given away, items received, estimated CO<sub>2</sub> saved
- Community-level metrics: total neighbourhood circulation
- Social motivation — not just data tracking, but a sense of shared purpose

## ONBOARDING

- Simple sign-up tied to postcode (Inner North Melbourne focus)
- Friendly, empathetic tone — "Start small. Share one thing."
  - Guided first post to reduce barrier to entry

# SYSTEM IMPACT

Unclutter intervenes at multiple levels of the system mapped in our research:

Measurable indicators (projected):

- Items redistributed per month per neighbourhood
- Reduction in hard-rubbish collection volume
- User-reported sense of community belonging (pre/post survey)
- CO<sub>2</sub> savings from item reuse vs. new purchase

## INDIVIDUAL LEVEL

- Reduces decision fatigue through simplified pathways (validated by Declutter Decision Kit testing)
- Replaces guilt-driven inaction with community-driven motivation
- Shifts emotional relationship with possessions: from shame to shared purpose

## COMMUNITY LEVEL

- Creates local redistribution infrastructure where none existed
- Builds social connection between neighbours — countering loneliness as a clutter driver (Jackson, 2017)
- Enables peer-to-peer circulation, reducing reliance on overwhelmed donation centres

## SYSTEMIC LEVEL

- Diverts usable goods from landfill and the exported waste stream
  - Reduces pressure on op-shops (currently unable to absorb 70% of donations — SBS, 2025)
- Models a community-owned alternative to commercial resale platforms
- Supports circular economy principles at the neighbourhood scale (Ellen MacArthur Foundation, 2013)

## REINFORCING LOOP INTERVENTION

(reference the system map from earlier slides)

Breaks the Clutter → Overwhelm → Avoidance → More Accumulation cycle

Introduces a new loop: Item posted → Neighbour receives → Connection made → Motivation to share more

Shifts from Storage as Solution → Circulation as Design

# LIMITATIONS & FUTURE DIRECTIONS

## LIMITATIONS

1. Small research sample — 6 semi-structured interviews in 3 Inner North suburbs (Fitzroy, Collingwood, Brunswick). Findings may not generalise to other demographics or geographies.
2. Prototype stage only — Unclutter remains an evocative prototype. No functional platform has been built or tested with real users at scale.
3. Geographic constraint — Designed specifically for Inner North Melbourne. Urban density, demographics, and existing infrastructure may differ significantly in suburban or regional contexts.
4. Self-reported data — Interview themes rely on participant recollection and stated preferences, which may differ from actual behaviour.
5. No longitudinal testing — We cannot yet measure sustained engagement or long-term behaviour change. Initial willingness does not guarantee ongoing participation.
6. Trust & safety considerations — Peer-to-peer exchanges between strangers raise safety, moderation, and quality-control challenges not yet addressed in the prototype.

## FUTURE DIRECTIONS

1. Expand research — Conduct a second round of interviews (n=20+) across broader Melbourne suburbs, including culturally diverse communities and different housing types.
2. Build a medium-fidelity prototype — Develop a clickable Figma prototype of the Give & Receive Feed, onboarding flow, and Local Hub Locator for structured usability testing.
3. Conduct usability testing — Test with 5–8 participants from target demographics. Measure task completion, emotional response, and perceived trust.
4. Map community partnerships — Approach local councils, op-shops, repair cafés, and community centres in the Inner North to assess real-world integration feasibility.
5. Research governance models — Investigate cooperative and community-owned platform structures to ensure Unclutter remains free, community-led, and independent from corporate interests.
6. Pilot the Circular Route Van — Test a small-scale mobile collection service in partnership with a local council to validate the logistical access hypothesis.

# REFLECTION & CLOSING STATEMENT

## WHAT WE LEARNED

This iterative process fundamentally changed how we understand clutter. Beginning with individual behaviour, our research and prototyping revealed that the real design opportunity lies in community infrastructure — not in reminding people to tidy up.

## CREATIVE CONFIDENCE

Each iteration gave us more confidence to challenge assumptions. The scrappy prototypes were wrong in the right ways — they exposed what really matters.

## SYSTEMS THINKING

We learned to trace problems to their root causes rather than symptoms. Clutter is a systemic failure, and design must respond at that level.

## COMMUNITY AS CORE

The consistent signal from research was belonging. The most effective solutions centre people connecting, not apps tracking.



*"The act of sharing is an act of community-building. Design that enables circulation enables belonging."*



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